Most of the transition elements are hexagonal close packed, cubic close packed or body centrered cubic which are the characteristics of true metals.

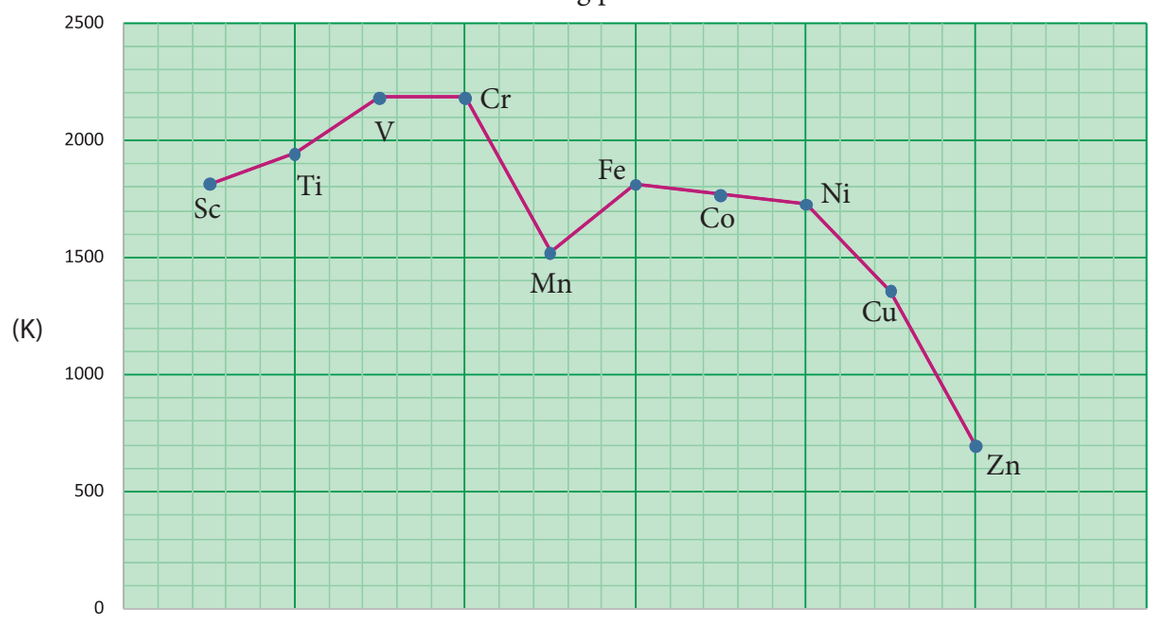
21 Sc HCP	22 Ti HCP	23 V BCC	24 Cr BCC	25 Mn BCC	26 Fe BCC	27 Co HCP	28 Ni FCC	29 Cu FCC	30 Zn HCP
39 Y HCP	40 Zr HCP	41 Nb BCC	42 Mo BCC	43 Tc HCP	44 Ru HCP	45 Rh FCC	46 Pd FCC	47 Ag FCC	48 Cd HCP
57* La DHCP	72 Hf HCP	73 Ta BCC/ TETR	74 W BCC	75 Re HCP	76 Os HCP	77 Ir FCC	78 Pt FCC	79 Au FCC	80 Hg RHO
89** Ac FCC	104 Rf [HCP]	105 Db [BCC]	106 Sg [BCC]	107 Bh [HCP]	108 Hs [HCP]	109 Mt [FCC]	110 Ds [BCC]	111 Rg [BCC]	112 Cn [BCC]

Figure 4.1 lattice structures of 3d, 4d and 5d transition metals

As we move from left to right along the transition metal series, melting point first increases as the number of unpaired d electrons available for metallic bonding increases, reach a maximum value and then decreases, as the d electrons pair up and become less available for bonding.

For example, in the first series the melting point increases from Scandium (m.pt 1814K) to a maximum of 2183 K for vanadium, which is close to 2180K for chromium. However, manganese in 3d series and Tc in 4d series have low melting point. The maximum melting point at about the middle of transition metal series indicates that d^5 configuration is favorable for strong interatomic attraction. The following figure shows the trends in melting points of transition elements.

Variation in melting point of 3d series elements



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